# Algebra 2 — Unit 1, Lesson 1.2

## Homework — Answer Key

Name: ____________________________ Date: ________________ Period: _________

### Practice

Every solution gets **checked**. Every inequality gets **graphed**. Every quadratic gets one side set to **zero** before anything else happens.

1. **Solve, and justify one step.** For each equation, give the solution and name one property of equality you used along the way.

| Equation | Solution | One property you used |
|---|---|---|
| (a) $4x - 9 = 3x + 2$ | $x = 11$ | **subtraction property of equality (subtract $3x$)** |
| (b) $6(x + 2) = 4x - 2$ | $x = -7$ | **distributive property, then division by 2** |
| (c) $\frac{x}{3} - 5 = 1$ | $x = 18$ | **multiplication property of equality (times 3)** |
| (d) $3(2x - 1) + 4 = 5x + 9$ | $x = 8$ | **subtraction property of equality (subtract $5x$)** |

Check your answer to (d) in the original equation. $3(16 - 1) + 4 = 45 + 4 = 49$ **and** $5(8) + 9 = 49$

2. **How many solutions?** You are given only the *last line* of each student's work.

(a) $7 = 7$ → **infinitely many** (b) $x = -2$ → **one**

(c) $4 = 9$ → **none** (d) $0 = 0$ → **infinitely many**

Which of the four came from an **identity**? **(a) and (d)** From a **contradiction**? **(c)**

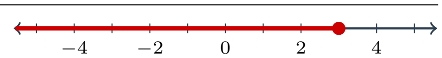
3. **Literal equations.** Solve each for the variable named.

| | | |
|---|---|---|
| (a) $V = \ell wh$ | solve for $w$ | $w = \dfrac{V}{\ell h}$ |
| (b) $2x + 5y = 20$ | solve for $y$ | $y = \dfrac{20 - 2x}{5} = 4 - \frac{2}{5}x$ |
| (c) $cx - 4x = d$ | solve for $x$ | $x = \dfrac{d}{c - 4}, \quad c \neq 4$ |

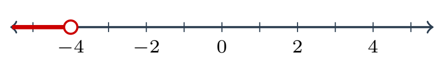
Item (c) is different from the other two. What did you have to do first, and why? **Factor $x$ out: $x(c - 4) = d$. The target variable appeared in two terms, and you cannot divide it away until it appears exactly once.** ....................................................................

4. **Solve and graph.** Shade each solution set. Open ∘ for $<$ and $>$, closed • for $\leq$ and $\geq$.

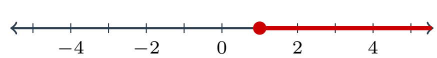
(a) $3x - 5 \leq 4$ $x \leq 3$



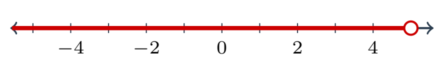

(b) $-2x + 1 > 9$ $x < -4$




(c) $5(x + 1) \geq 3x + 7$ $x \geq 1$




(d) $2(x - 1) < x + 3$ $x < 5$




Exactly one of those four made you reverse the symbol. Which, and at which step? **(b), when dividing both sides of $-2x > 8$ by $-2$**

5. **Write it, solve it, say it.** Answer each in a sentence with units — not as "$x =$ something."

   (a) A gym charges a \$25 sign-up fee plus \$12 per month. Ana has paid \$121 in total.
   Equation: $25 + 12m = 121$  Answer: **Ana has been a member for 8 months.** . . . . . . . . . . . . . . . . . .

   (b) A club orders T-shirts: a \$45 setup fee plus \$8 per shirt, and the budget is **at most** \$300.
   Inequality: $45 + 8s \le 300$  Answer: $s \le 31.875$, **so the club can order at most 31 shirts — shirts come in whole numbers, so the decimal has to be rounded *down*, not to the nearest.**

6. **Quadratics — pick your method.** Two are square-root problems, two factor, and two need the formula (leave those in simplest radical form).

   (a) $x^2 = 64, \quad x = \pm 8$  (b) $3x^2 - 27 = 0, \quad x = \pm 3$

   (c) $x^2 - 2x - 24 = 0, \quad x = 6,\ -4$  (d) $x^2 + 7x = -10, \quad x = -5,\ -2$

   (e) $x^2 - 6x + 2 = 0, \quad x = 3 \pm \sqrt{7}$  (f) $2x^2 + 4x - 1 = 0, \quad x = \dfrac{-2 \pm \sqrt{6}}{2}$

   **The two that need the formula — work them in full.**

   (e) $x^2 - 6x + 2 = 0$

$$x = \frac{6 \pm \sqrt{(-6)^2 - 4(1)(2)}}{2(1)} = \frac{6 \pm \sqrt{28}}{2} = \frac{6 \pm 2\sqrt{7}}{2} = 3 \pm \sqrt{7}$$

   (f) $2x^2 + 4x - 1 = 0$

$$x = \frac{-4 \pm \sqrt{4^2 - 4(2)(-1)}}{2(2)} = \frac{-4 \pm \sqrt{24}}{4} = \frac{-4 \pm 2\sqrt{6}}{4} = \frac{-2 \pm \sqrt{6}}{2}$$

   Which item required a Step 0 the others did not, and what was it? **(d) — add 10 to both sides first so one side is 0**
   Verify one solution of (c) by substituting it back. **At $x = 6$: $36 - 12 - 24 = 0$.** **(Or at $x = -4$: $16 + 8 - 24 = 0$.)**

7. **Count before you solve.** Fill the table in *without* solving anything.

| **Equation** | $b^2 - 4ac$ | **Positive / zero / negative?** | **# real solutions** |
|---|---|---|---|
| (a) $x^2 + 5x + 6 = 0$ | $1$ | **positive** | **two** |
| (b) $x^2 + 4x + 4 = 0$ | $0$ | **zero** | **one** |
| (c) $x^2 + x + 5 = 0$ | $-19$ | **negative** | **none** |
| (d) $3x^2 - 2x - 1 = 0$ | $16$ | **positive** | **two** |

One of those four has a graph that touches the $x$-axis without crossing it. Which? **(b)**

8. **Explain.** A classmate solves $(x+1)(x-4)=6$ by writing "$x+1=6$ or $x-4=6$." Explain what is wrong, why the zero product property needs the zero, and then solve the equation correctly. **The zero product property applies only when the product is *zero*. Six can be written as a product of two nonzero numbers in infinitely many ways ($6\cdot 1$, $3\cdot 2$, $12\cdot \frac{1}{2}$, ...), so knowing the product is 6 forces nothing about either factor. Zero is the only number that cannot be built from two nonzero factors — that uniqueness is the whole property.**

$$\begin{aligned}(x+1)(x-4)&=6\\ x^2-3x-4&=6\\ x^2-3x-10&=0\\ (x-5)(x+2)&=0\\ x=5 \quad &\text{or} \quad x=-2\end{aligned}$$

**Extension** optional

**The solution-count audit.** In each problem below, a letter sits *inside* the equation. Your job is to say which values of that letter produce each possible number of solutions — so the count becomes something you control rather than something you discover.

| **Number of solutions** | **For which $c$ does $5x+c=5x+8$ ?** | **For which $b$ does $x^2+bx+9=0$ have that many *real* solutions?** |
|---|---|---|
| two | — | $b<-6$ **or** $b>6$ |
| exactly one | **no such** $c$ | $b=6$ **or** $b=-6$ |
| none | **every** $c\neq 8$ | $-6<b<6$ |
| infinitely many | $c=8$ | — |

(1) One box in the left column is impossible to fill. Which, and why can that never happen for $5x+c=5x+8$? **"Exactly one." The $5x$ terms are identical on both sides, so they always cancel — the variable can never survive, and a surviving variable is the only way to get exactly one solution.**

(2) A linear equation *can* have infinitely many solutions. A quadratic equation with $a\neq 0$ never can. Explain why not — what is it about the $x^2$ term that makes an identity impossible? **For an identity, everything containing the variable has to cancel. In a quadratic with $a\neq 0$ the $ax^2$ term cannot be cancelled by anything else present, so the variable never fully disappears — and the quadratic formula then hands back at most two values, because the only choice in it is the $\pm$.**

(3) In the right column, the answers to "two" and "none" are ranges of $b$, not single values. Say what that means about the graph of $y=x^2+bx+9$ as $b$ increases from 0 through 6 and past it. **At $b=0$ the parabola $y=x^2+9$ sits entirely above the $x$-axis — no real zeros. As $b$ grows, the vertex $\left(-\frac{b}{2},\ 9-\frac{b^2}{4}\right)$ slides left and *down*. At exactly $b=6$ the vertex lands on the axis at $(-3,0)$ and the graph touches without crossing — one solution. Past $b=6$ the vertex is below the axis and the graph cuts it twice. The discriminant is just an algebraic way of asking how far down that vertex has gone.**

**Connections & Big Ideas**

**Coming next — Lesson 1.3: Functions.** The third and last Algebra 1 domain area, and today's equations become its machinery. A function's **zeros** are defined as the solutions of $f(x) = 0$ — so every technique in problem 6 is a way of finding where a graph crosses the $x$-axis, and the discriminant in problem 7 is a way of counting those crossings before you draw anything. The inequalities in problem 4 are how a function's **domain** gets written down when some inputs are not allowed. And the literal equations in problem 3 are the reason $y = mx + b$ has three different "forms" in Unit 2: it is one equation, solved for different things.

**Teacher Note**

**Grade problem 2 first.** It takes thirty seconds per paper and it separates students who can *interpret* a result from students who can only compute one. A student who is fluent everywhere else but writes "one solution" for $7 = 7$ needs one sentence of correction, not a reteach.

**Problem 6(d) is the one that carries.** $x^2 + 7x = -10$ is the only item on the page whose first move is not the obvious one, and a student who attacks it as $x(x + 7) = -10$ will produce confident nonsense. Pair it with problem 8, which asks for the same idea in words — if a student gets 6(d) right but 8 wrong, they have the procedure without the reason and will lose it by Unit 3.

**Problem 5(b) rewards reading, not algebra.** $s \leq 31.875$ becomes "31 shirts," and rounding to 32 blows the budget. **A.EI.1f** asks students to *interpret* solutions in context, and this is where that gets graded — accept no answer that stops at the decimal.

**The Extension is optional but item (3) is the best question on the page.** It turns the discriminant into a statement about a vertex sliding down through the $x$-axis, which is exactly the Unit 3 picture. If any student writes that connection unprompted, put it on the board.